



EEE240 FUNDAMENTALS OF DIGITAL LOGIC DESIGN

Chapter no. 1

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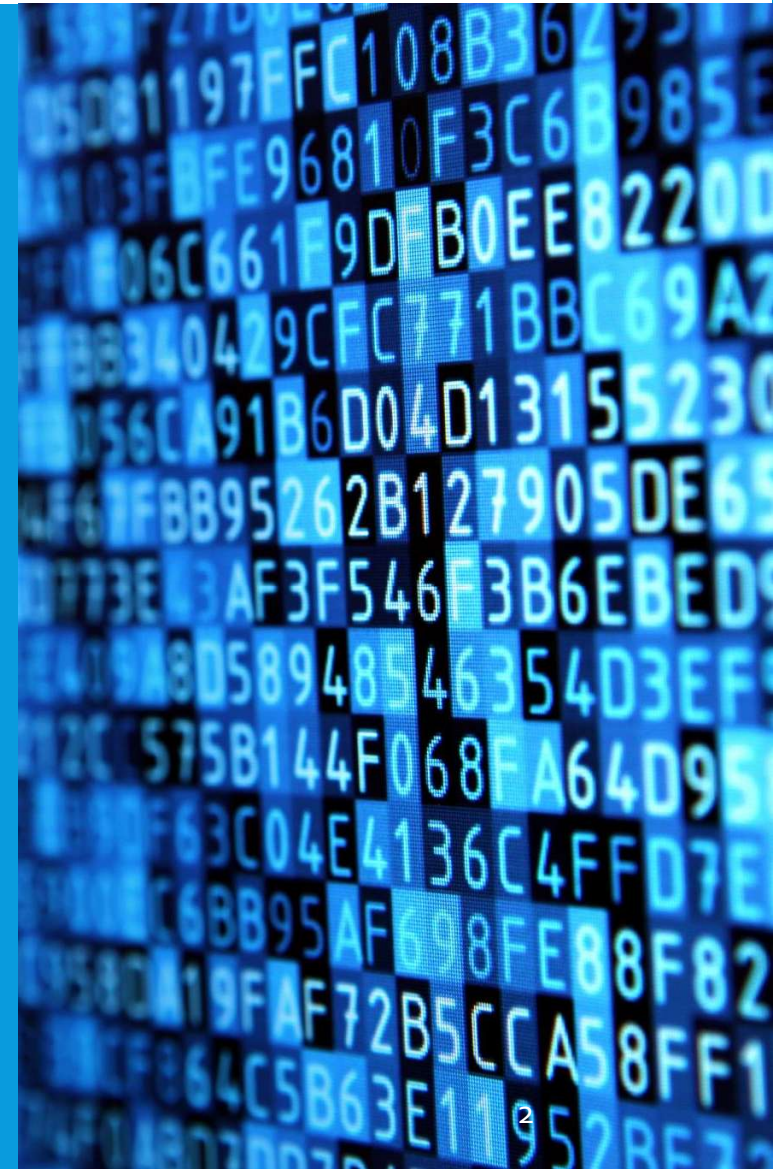
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OUTLINE

- Digital Systems
- Binary Numbers
- Number-Base Conversions
- Octal and Hexadecimal Numbers
- Complements of Numbers
- Subtraction with complements
- Signed Binary Numbers
- Binary Codes
- Binary Storage and Registers
- Binary Logic



MUST READ!!!

- Chapter No. 1: Digital Systems and Binary Numbers

Digital Systems

- Discrete Data

- Examples:

- 26 letters of the alphabet (A, B ... etc.)
 - 10 decimal digits (0, 1, 2 ... etc.)

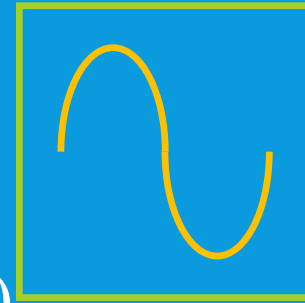
- Combine together

- Words are made of letters (University ... etc.)
 - Numbers are made of digits (4241 ... etc.)

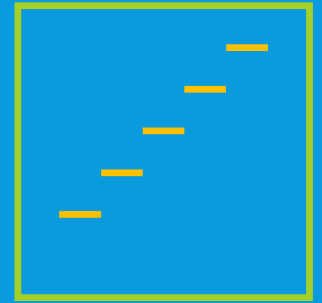
- Binary System

- Only '0' and '1' digits

- Can be easily implemented in electronic circuits



Analog



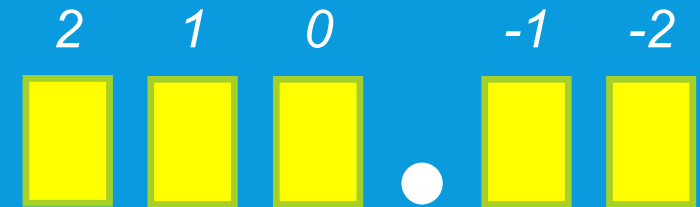
Discrete

Decimal Number System

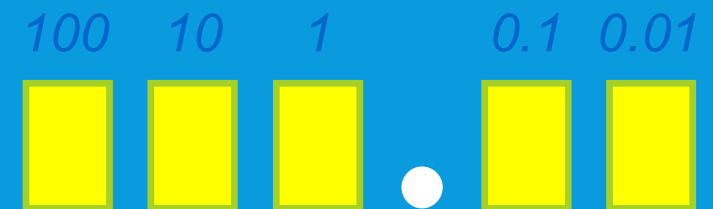
- Base (also called radix) = 10
 - 10 digits { 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 }



- Digit Position
 - Integer & fraction



- Digit Weight
 - Weight = $(Base)^{Position}$



- Magnitude
 - Sum of “*Digit x Weight*”

500 10 2 0.7 0.04

$$d_2 * B^2 + d_1 * B^1 + d_0 * B^0 + d_{-1} * B^{-1} + d_{-2} * B^{-2}$$

- Formal Notation

$$(512.74)_{10}$$

BINARY SYSTEMS

- Review, how the decimal system is designed?
 - Base (Radix) 10
 - Symbols: 0, 1, 2, ... 8, 9
 - What if quantity greater than 9?
 - Add more significant digits to the left
 - Each position carries a weight
 - Example: 7392
 - $7 \times 10^3 + 3 \times 10^2 + 9 \times 10^1 + 2 \times 10^0 = 7392$
 - Seven thousand three hundred and ninety two
 - Weightage increases as the digit placement moves to left and vice versa

BINARY SYSTEMS

What if weightage is less than 10^0

- We use the symbol point (.) and write the number to the right
- **MSD** $\rightarrow$

10^3	10^2	10^1	10^0	.	10^{-1}	10^{-2}
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 $\leftarrow$ **LSD**
- Base-10 is default base for numbering system
- But, we can generalize the rule for any base-**r** system from the base-10 (decimal) number system
 - Base (Radix) is “**r**”
 - Symbols: **0** $\rightarrow$ (**r** – 1)
 - Weightage of each number increases by a factor of “**r**” as its placement moves to the left

r^3	r^2	r^1	r^0	.	r^{-1}	r^{-2}
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BINARY SYSTEMS

- Example: $a_n \dots a_5 a_4 a_3 a_2 a_1 a_0 \bullet a_{-1} a_{-2} a_{-3} \dots a_{-m}$

$$a_n \times r^n + \dots a_5 \times r^5 + a_4 \times r^4 + a_3 \times r^3 + a_2 \times r^2 + a_1 \times r^1 + a_0 \times r^0 \\ + a_{-1} \times r^{-1} + a_{-2} \times r^{-2} + a_{-3} \times r^{-3} + \dots a_{-m} \times r^{-m}$$

- Simplified:
 - $a_j = \{0 \rightarrow (r - 1)\}$
 - j is the location of digit “ a_j ” of the quantity in decimal numbers is:

$$D = \sum_{j=-m}^n a_j \times r^j$$

BINARY SYSTEMS

- As we know the general rule, now we can understand how the binary (base-2) would work
 - Base (Radix): “2”
 - Symbols: 0, 1
 - Weightage of each number increases by a factor of “2” as its placement moves to the left
- Example:
 - $(11010)_2$ (Don’t call it eleven thousand and ten, **but** one, one, zero, one, zero; we are not in decimal system)
 - $(11010.11)_2$

SPECIAL POWERS OF 2

2^{10} (=1024) is **Kilo**, denoted "**K**"

2^{20} (=1,048,576) is **Mega**, denoted "**M**"

2^{30} (1,073, 741,824) is **Giga**, denoted "**G**"

2^{40} (1,099,511,627,776) is **Tera**, denoted "**T**"

Note: 8 bits (b) are also called a byte (B)

Exercise: what is $(1111\ 1111)_2$ equal to in decimal?

Also, what is the maximum number (in decimal) a two bytes long word can represent?

RANGE OF BINARY NUMBERS

- 000, 001 ... ,111 or in decimal system: 0, 1 ... 7
 - Total of 8 numbers ($=2^3$)
 - Range: from 0 to 7 (0 to $2^3 - 1$)
- In general an n-bit number represents:
 - 2^n different numbers
 - Min: 0
 - Max number: $2^n - 1$
- For fractions: m bits after the radix point:
 - Min: 0
 - Max number: $(2^m - 1)/2^m$

The Power of 2

n	2^n
0	$2^0=1$
1	$2^1=2$
2	$2^2=4$
3	$2^3=8$
4	$2^4=16$
5	$2^5=32$
6	$2^6=64$
7	$2^7=128$



n	2^n
8	$2^8=256$
9	$2^9=512$
10	$2^{10}=1024$
11	$2^{11}=2048$
12	$2^{12}=4096$
20	$2^{20}=1M$
30	$2^{30}=1G$
40	$2^{40}=1T$

Kilo

Mega

Giga

Tera

EXAMPLES OF BINARY NUMBERS

- 16 Gbyte (16 GB) of memory
 - How many (address) bits are required to address each byte?
- Digital camera has 2048x2048 pixels, and each pixel stores 8 bits of information:
 - a. How many Mega pixels?
 - b. How many bits are stored per frame?
 - c. How many different intensity levels can be represented by each pixel?

BASE-5 SYSTEMS

- Quinary System
 - Base (Radix): “5”
 - Symbols: 0, 1, 2, 3, 4
 - Weightage of each number increases by a factor of “5” as its placement moves to the left
- Example:
 - $(4021.2)_5$ (Don’t call it four thousand and twenty one, **but** four, zero, two, one, point two)

BASE-8 SYSTEMS

- Octal System
 - Base (Radix): “8”
 - Symbols: 0, 1, 2, ... 6, 7
 - Weightage of each number increases by a factor of “8” as its placement moves to the left
- Example:
 - $(127.4)_8$ (Don’t call it one twenty seven point four, **but** one, two, seven, point four)

BASE-16 SYSTEMS

- Hexadecimal System
 - Base (Radix): “16”
 - Symbols: 0, 1, 2, ... 8, 9, A, B, C, D, E, F
 - Weightage of each number increases by a factor of “16” as its placement moves to the left
- Example:
 - $(B65F)_{16}$
- Hexadecimal numbers are used to represent long strings of bits in addresses, instructions and data

CONVERSION BETWEEN NUMBER SYSTEMS

- Conversion of decimal integer
 - Repeated division
 - Examples:
 - 57
 - Binary
 - Quinary
 - Octal
 - Hexadecimal

$57/2 = 28: \text{ rem}=1=a_0$
 $28/2 = 14: \text{ rem}=0=a_1$
 $14/2 = 7: \text{ rem}=0=a_2$
 $7/2 = 3: \text{ rem}=1=a_3$
 $3/2 = 1: \text{ rem}=1=a_4$
 $1/2 = 0: \text{ rem}=1=a_5$
 $(111001)_2$

$57/5 = 11: \text{ rem}=2$
 $11/5 = 2: \text{ rem}=1$
 $(212)_5$

$57/8 = 7: \text{ rem}=1$
 $(71)_8$

$57/16 = 3: \text{ rem}=9$
 $(39)_{16}$

CONVERSION BETWEEN NUMBER SYSTEMS

- Conversion of decimal fraction

- Repeated multiplication

- Examples:

- 0.625

- Binary
 - Quinary
 - Octal
 - Hexadecimal

$$\begin{aligned} 0.625 \times 2 &= 1.250: & a_{-1} &= 1 \\ 0.250 \times 2 &= 0.50: & a_{-2} &= 0 \\ 0.500 \times 2 &= 1.0: & a_{-3} &= 1 \\ & & (0.101)_2 & \end{aligned}$$

$$\begin{aligned} 0.625 \times 5 &= 3.125: & a_{-1} &= 3 \\ 0.125 \times 5 &= 0.625: & a_{-2} &= 0 \\ 0.625 \times 5 &= 3.125: & a_{-3} &= 3 \\ & & (0.303)_5 & \end{aligned}$$

$$\begin{aligned} 0.625 \times 8 &= 5.0: & a_{-1} &= 5 \\ 0.0 \times 8 &= 0.0: & a_{-2} &= 0 \\ & & (0.50)_8 & \end{aligned}$$

$$\begin{aligned} 0.625 \times 16 &= 10(A): & a_{-1} &= A \\ 0.0 \times 16 &= 0.0: & a_{-2} &= 0 \\ & & (0.A0)_{16} & \end{aligned}$$

CONVERSION BETWEEN NUMBER SYSTEMS

- Conversion of mixed decimal numbers
 - Convert the integer part using repeated division
 - Convert the fraction part using repeated multiplication
 - Combine the integer and fractional components in the new base
- Example:
 - $(48.5625)_{10}$

CONVERSION BETWEEN NUMBER SYSTEMS

- Conversion to binary number system:

- $(48.5625)_{10} = (?)_2$
- Repeated division

$48/2 = 24: a_0 = 0$
 $24/2 = 12: a_1 = 0$
 $12/2 = 6: a_2 = 0$
 $6/2 = 3: a_3 = 0$
 $3/2 = 1: a_4 = 1$
 $1/2 = 0: a_5 = 1$
 $(110000)_2$

Repeated multiplication

$0.5625 \times 2 = 1.1250: a_{-1} = 1$
 $0.1250 \times 2 = 0.2500: a_{-2} = 0$
 $0.2500 \times 2 = 0.5000: a_{-3} = 0$
 $0.5000 \times 2 = 1.0000: a_{-4} = 1$
 $(0.1001)_2$

- $(48.5625)_{10} = (110000.1001)_2$

CONVERSION BETWEEN NUMBER SYSTEMS

- Conversion to quinary number system:

- $(48.5625)_{10} = (?)_5$

- Repeated division

$$\begin{aligned} 48/5 &= 9: a_0 = 3 \\ 9/5 &= 1: a_1 = 4 \\ 1/5 &= 0: a_2 = 1 \\ \mathbf{(143)_5} \end{aligned}$$

Repeated multiplication

$$\begin{aligned} 0.5625 \times 5 &= 2.8125: a_{-1} = 2 \\ 0.8125 \times 5 &= 4.0625: a_{-2} = 4 \\ 0.0625 \times 5 &= 0.3125: a_{-3} = 0 \\ 0.3125 \times 5 &= 1.5625: a_{-4} = 1 \\ \mathbf{(0.2401)_5} \end{aligned}$$

- $(48.5625)_{10} = (143.2401)_5$

CONVERSION BETWEEN NUMBER SYSTEMS

- Conversion to octal number system:

- $(48.5625)_{10} = (?)_8$

- Repeated division

$$\begin{aligned} 48/8 &= 6: a_0 = 0 \\ 6/8 &= 0: a_1 = 6 \\ (60)_8 \end{aligned}$$

Repeated multiplication

$$\begin{aligned} 0.5625 \times 8 &= 4.5000: a_{-1} = 4 \\ 0.5000 \times 8 &= 4.0000: a_{-2} = 4 \\ 0.0000 \times 8 &= 0.0000: a_{-3} = 0 \\ (0.440)_8 \end{aligned}$$

- $(48.5625)_{10} = (60.44)_8$

CONVERSION BETWEEN NUMBER SYSTEMS

- Conversion to hexadecimal number system:

- $(48.5625)_{10} = (?)_{16}$

- Repeated division

$$\begin{aligned} 48/16 &= 3: a_0 = 0 \\ 3/16 &= 0: a_1 = 3 \\ \mathbf{(30)}_{16} \end{aligned}$$

Repeated multiplication

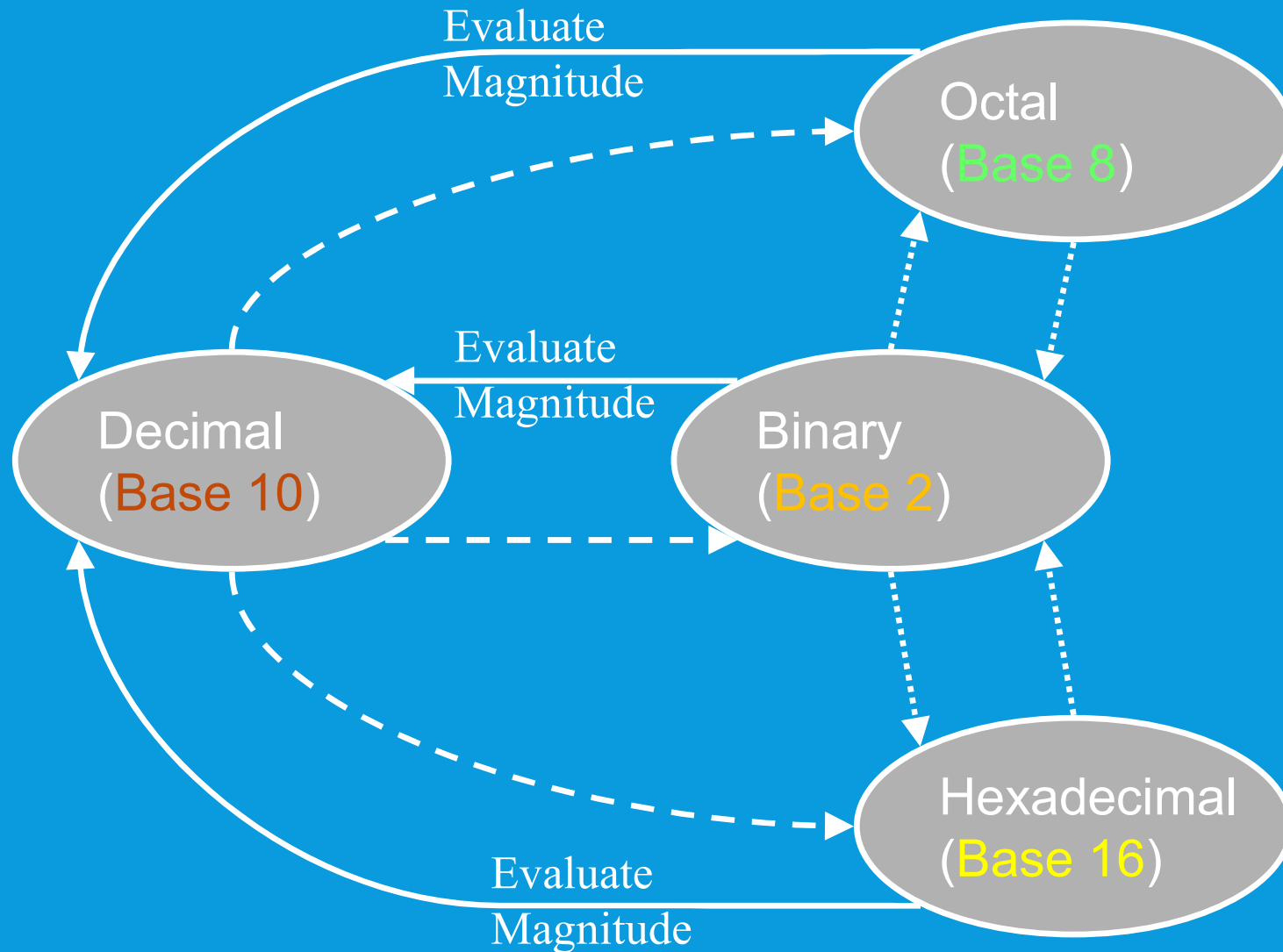
$$\begin{aligned} 0.5625 \times 16 &= 9.000: a_{-1} = 9 \\ 0.0000 \times 16 &= 0.000: a_{-2} = 0 \\ \mathbf{(0.90)}_{16} \end{aligned}$$

- $(48.5625)_{10} = (30.90)_{16}$

CONVERSION BETWEEN NUMBER SYSTEMS

- Conversion from base-A to base-B can be carried out by repeated division and repeated multiplication
 - Make sure you do the division and multiplication using the correct base
 - $\text{Base-A} \rightarrow \text{Base-B}$
- It is easier (generally) to convert base-A to base-10 and then to base-B
 - $\text{Base-A} \rightarrow \text{Base-10} \rightarrow \text{Base-B}$

Number Base Conversions



CONVERSION BETWEEN NUMBER SYSTEMS

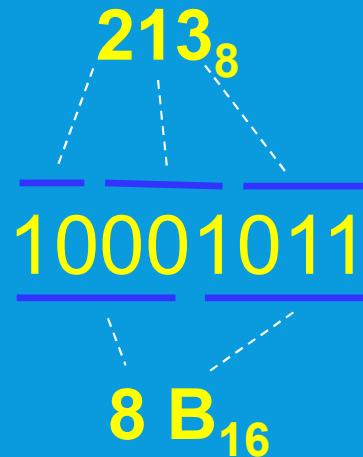
- Conversion between **binary and octal**
 - Each octal digit corresponds to three (3) bits
 - Examples:
 - $(101\ 110\ 010\ .\ 011\ 001)_2 = (562.31)_8$
 - $(010\ 011\ 100\ .\ 101\ 001)_2 = (234.51)_8$
 - $(745.32)_8 = (111\ 100\ 101\ .\ 011\ 010)_2$
 - $(306.05)_8 = (011\ 000\ 110\ .\ 000\ 101)_2$

CONVERSION BETWEEN NUMBER SYSTEMS

- Conversion between **binary and hexadecimal**
 - Each hexadecimal digit corresponds to four(4) bits
 - Examples:
 - $(1001\ 1010\ 0110\ .\ 1011\ 0101)_2 = (9A6.B5)_{16}$
 - $(1100\ 1011\ 1000\ .\ 1110\ 0111)_2 = (CB8.E7)_{16}$
 - $(E94.D2)_{16} = (1110\ 1001\ 0100\ .\ 1101\ 0010)_2$
 - $(1C7.8F)_{16} = (0001\ 1100\ 0111\ .\ 1000\ 1111)_2$

CONVERSION BETWEEN NUMBER SYSTEMS

- Conversion octal to hexadecimal and vice versa using binary
 - Convert **213₈** into a hexadecimal number:



Decimal	Binary	Octal	Hex
00	0000	00	0
01	0001	01	1
02	0010	02	2
03	0011	03	3
04	0100	04	4
05	0101	05	5
06	0110	06	6
07	0111	07	7
08	1000	10	8
09	1001	11	9
10	1010	12	A
11	1011	13	B
12	1100	14	C
13	1101	15	D
14	1110	16	E
15	1111	17	F

DECIMAL, BINARY, OCTAL AND HEXADECIMAL

ARITHMETIC

- Decimal Addition

$$\begin{array}{r} 1 \quad 1 \\ 5 \quad 5 \\ + \quad 5 \quad 5 \\ \hline 1 \quad 1 \quad 0 \end{array} \quad \leftarrow \text{Carry}$$

$= \text{Ten} \geq \text{Base}$

BINARY ARITHMETIC

$$\begin{array}{r} \\ \\ + \\ \hline \\ \hline \end{array}$$

← augend

← addend

Binary Addition

- Column Addition

$$\begin{array}{rcccccc} & 1 & 1 & 1 & 1 & 1 & 1 & & \\ & & 1 & 1 & 1 & 1 & 0 & 1 & = 61 \\ + & & & 1 & 0 & 1 & 1 & 1 & = 23 \\ \hline & 1 & 0 & 1 & 0 & 1 & 0 & 0 & = 84 \end{array}$$

 $\geq (2)_{10}$

BINARY ARITHMETIC

$$\begin{array}{r} 101101 \\ - 100111 \\ \hline 000110 \end{array}$$

← minuend
← subtrahend

Binary Subtraction

- Borrow a “Base” when needed

$$\begin{array}{r}
 \begin{array}{ccccccc}
 & & 1 & & & 2 & \\
 0 & \cancel{2} & 2 & 0 & 0 & 2 & \\
 \cancel{1} & 0 & 0 & \cancel{1} & \cancel{1} & 0 & 1 \\
 - & & & 1 & 0 & 1 & 1 & 1 \\
 \hline
 0 & 1 & 1 & 0 & 1 & 1 & 0
 \end{array}
 & \begin{array}{l}
 = (10)_2 \\
 \\
 = 77 \\
 = 23 \\
 \\
 = 54
 \end{array}
 \end{array}$$

Note: The diagram illustrates binary subtraction with borrowing. The top row shows the minuend (101010) and the subtrahend (010111). The result is 0110110. The borrowing process is indicated by the orange arrows and the '2' values above the digits.

BINARY ARITHMETIC

$$\begin{array}{r} \\ \\ \hline \\ \\ 1011 \\ \hline 110111 \\ \hline \end{array}$$

← multiplicand
← multiplier

Binary Multiplication

- Bit by bit

$$\begin{array}{r} 1 0 1 1 1 \\ x 1 0 1 0 \\ \hline 0 0 0 0 0 \\ 1 0 1 1 1 \\ 0 0 0 0 0 \\ 1 0 1 1 1 \\ \hline 1 1 1 0 0 1 1 0 \end{array}$$

BINARY DIVISION (EXAMPLE 1)

Dividend

Divisor

↓ ↓
1 0 0 0 0 0 1 / 1 1 0 0 = ?

1 1 0 0 | 1 0 0 0 0 0 1

← Quotient

← shift, if divisor is bigger and write 1 in quotient

BINARY DIVISION (EXAMPLE 1)

Dividend

Divisor

↓ ↓
1 0 0 0 0 0 1 / 1 1 0 0 = ?

0 1
1 1 0 0 | 1 0 0 0 0 0 1
1 1 0 0
0 1 0 0

← Quotient

← shift, if divisor is bigger and write 1 in quotient

BINARY DIVISION (EXAMPLE 1)

Dividend

Divisor

$$\begin{array}{c} \downarrow \\ 1\ 0\ 0\ 0\ 0\ 0\ 1 \end{array} / \begin{array}{c} \downarrow \\ 1\ 1\ 0\ 0 \end{array} = ?$$

$$\begin{array}{r} 1 \\ 1\ 1\ 0\ 0 \overline{) 1\ 0\ 0\ 0\ 0\ 0\ 1} \\ \underline{1\ 1\ 0\ 0} \downarrow \\ 0\ 1\ 0\ 0\ 0 \end{array}$$

← Quotient

← shift, if divisor is bigger and write 1 in quotient

← subtract, and bring next digit

BINARY DIVISION (EXAMPLE 1)

Dividend

Divisor

$$\begin{array}{c} \downarrow \\ 1000001 / 1100 = ? \end{array}$$

$$\begin{array}{r} 10 \\ 1100 \overline{) 1000001} \\ \underline{1100} \\ 0100001 \end{array}$$

← Quotient

← shift, if divisor is bigger and write 1 in quotient

← subtract, and bring next digit (bring again next digit)

BINARY DIVISION (EXAMPLE 1)

Dividend

Divisor

$$\begin{array}{c} \downarrow \\ 1\ 0\ 0\ 0\ 0\ 0\ 1 \end{array} / \begin{array}{c} \downarrow \\ 1\ 1\ 0\ 0 \end{array} = ?$$

$$\begin{array}{r} \underline{1\ 0\ 1} \\ 1\ 1\ 0\ 0\ |\ 1\ 0\ 0\ 0\ 0\ 0\ 1 \\ \underline{1\ 1\ 0\ 0} \downarrow \downarrow \\ 0\ 1\ 0\ 0\ 0\ 1 \\ \underline{1\ 1\ 0\ 0} \\ 0\ 1\ 0\ 1 \end{array}$$

← Quotient

← shift, if divisor is bigger and write 1 in quotient

← subtract, and bring next digit (bring again next digit)

← subtract

← Remainder

BINARY DIVISION (EXAMPLE 2)

Dividend

Divisor

$$\begin{array}{c} \downarrow \qquad \downarrow \\ 1\ 1\ 1\ 0\ 1\ 1\ / \ 1\ 1 = ? \end{array}$$

$$\begin{array}{r} 1 \\ \hline 1\ 1\ |\ 1\ 1\ 1\ 0\ 1\ 1 \\ \underline{1\ 1} \end{array}$$

← Quotient

← write 1 in quotient

BINARY DIVISION (EXAMPLE 2)

Dividend

Divisor

$$\begin{array}{c} \downarrow \\ 111011 / 11 = ? \end{array}$$

$$\begin{array}{r} 1 \\ \hline 11 \overline{) 111011} \\ \underline{11} \\ 001 \end{array}$$

← Quotient

← write 1 in quotient

← subtract, bring next digit

BINARY DIVISION (EXAMPLE 2)

Dividend

Divisor

$$\begin{array}{c} \downarrow \qquad \downarrow \\ 1\ 1\ 1\ 0\ 1\ 1\ / \ 1\ 1 = ? \end{array}$$

$$\begin{array}{r} 1\ 0 \\ \hline 1\ 1\ |\ 1\ 1\ 1\ 0\ 1\ 1 \\ \underline{1\ 1\ \downarrow\ \downarrow} \\ 0\ 0\ 1\ 0 \end{array}$$

← Quotient

← write 1 in quotient

← subtract, bring next digit (again)

BINARY DIVISION (EXAMPLE 2)

Dividend

Divisor

$$\begin{array}{c} \downarrow \\ 1\ 1\ 1\ 0\ 1\ 1 \end{array} / \begin{array}{c} \downarrow \\ 1\ 1 \end{array} = ?$$

$$\begin{array}{r} 1\ 0\ 0 \\ \hline 1\ 1 \mid 1\ 1\ 1\ 0\ 1\ 1 \\ \underline{1\ 1} \\ 0\ 0\ 1\ 0\ 1 \end{array}$$

← Quotient

← write 1 in quotient

← subtract, bring next digit (again) (again)

BINARY DIVISION (EXAMPLE 2)

Dividend

Divisor

$$\begin{array}{c} \downarrow \\ 111011 / 11 = ? \end{array}$$

$$\begin{array}{r} 1001 \\ 11 \overline{) 111011} \\ \underline{11} \\ 00101 \\ \underline{11} \\ 1 \end{array}$$

← Quotient

← write 1 in quotient

← subtract, bring next digit (again) (again)

← write 1 in quotient

BINARY DIVISION (EXAMPLE 2)

Dividend

Divisor

$$\begin{array}{c} \downarrow \\ 111011 / 11 = ? \end{array}$$

$$\begin{array}{r} 1001 \\ 11 \overline{) 111011} \end{array} \quad \leftarrow \text{Quotient}$$

$$\begin{array}{r} 11 \downarrow \downarrow \downarrow \\ \hline 00101 \end{array}$$

← write 1 in quotient

← subtract, bring next digit (again) (again)

$$\begin{array}{r} 11 \downarrow \\ \hline 101 \end{array}$$

← write 1 in quotient

← subtract, bring next digit

BINARY DIVISION (EXAMPLE 2)

Dividend

Divisor

$$\begin{array}{c} \downarrow \\ 111011 / 11 = ? \end{array}$$

$$\begin{array}{r} 10011 \\ 11 \overline{) 111011} \\ \underline{11} \\ 00101 \\ \underline{11} \\ 101 \\ \underline{11} \\ 0 \end{array} \quad \leftarrow \text{Quotient}$$

← write 1 in quotient

← subtract, bring next digit (again) (again)

← write 1 in quotient

← subtract, bring next digit

← write 1 in quotient

BINARY DIVISION (EXAMPLE 2)

Dividend

Divisor

$$\begin{array}{c} \downarrow \\ 111011 / 11 = ? \end{array}$$

$$\begin{array}{r} 10011 \\ 11 \overline{) 111011} \end{array} \quad \leftarrow \text{Quotient}$$

$$\begin{array}{r} 11 \downarrow \downarrow \downarrow \\ 00101 \end{array}$$

← write 1 in quotient

← subtract, bring next digit (again) (again)

$$\begin{array}{r} 11 \downarrow \\ 101 \end{array}$$

← write 1 in quotient

$$\begin{array}{r} 11 \\ 10 \end{array}$$

← subtract, bring next digit

$$\begin{array}{r} 11 \\ 10 \end{array}$$

← write 1 in quotient

$$10$$

← remainder

COMPLEMENTS OF NUMBERS

Complements are used to simplify subtraction operation

- Diminished Radix Complement
 - Given an “n” digits number “N” in base “r” the (r-1) complement is:
 - $(r^n - 1) - N$
 - Examples:

$$\begin{aligned} 9\text{'s complement of } 546700 &= (10^6 - 1) - 546700 \\ &= 999999 - 546700 \\ &= 453299 \end{aligned}$$

$$\begin{aligned} 9\text{'s complement of } 012398 &= (10^6 - 1) - 012398 \\ &= 999999 - 012398 \\ &= 987601 \end{aligned}$$

i.e. each digit is subtracted from 9

COMPLEMENTS OF NUMBERS

- For binary numbers:
 - $r = 2$ and $(r - 1) = 1$
 - For “n” = 4
 - $2^4 = (10000)_2$
 - $2^4 - 1 = (1111)_2$
- So, 1’s complement is
 - $2^4 - 1 - N = 1111 - N$
i.e. each digit is subtracted from 1
- However:
 - $1 - 1 = 0$ and $1 - 0 = 1$
- So, 1’s complement of binary numbers are formed by changing 1’s to 0’s and 0’s to 1’s
 - Examples:
 - 1’s complement of 1011000 = 0100111
 - 1’s complement of 0101101 = 1010010

COMPLEMENTS OF NUMBERS

- Radix Complement

- Given an “n” digits number “N” in base “r” the “r’s” complement is:

$$\begin{cases} r^n - N; N \neq 0 \\ 0; N = 0 \end{cases}$$

- Examples: for a six-digit decimal number (i.e. n=6)

$$\text{10's complement of 012398} = (10^6) - 012398 = 1000000 - 012398 = 987602$$

$$\text{10's complement of 246700} = (10^6) - 246700 = 1000000 - 246700 = 753300$$

- However, r’s complement is (r-1)’s complement + 1
- Obtaining (r-1)’s complement is easy
 - each digit is subtracted from 9

$$\begin{aligned} \text{10's complement of 012398} &= (999999 - 012398) + 1 \\ &= 987601 + 1 = 987602 \end{aligned}$$

$$\begin{aligned} \text{10's complement of 246700} &= (999999 - 246700) + 1 \\ &= 753299 + 1 = 753300 \end{aligned}$$

COMPLEMENTS OF NUMBERS

- 2's complement is obtained by adding 1 to 1's complement
- Example: 7 digits number (n=7)

$$\begin{aligned} 2's \text{ complement of } 1101100 &= 0010011 + 1 \\ &= 0010100 \end{aligned}$$

$$\begin{aligned} 2's \text{ complement of } 0110111 &= 1001000 + 1 \\ &= 1001001 \end{aligned}$$

i.e. The 2's complement is obtained by leaving the least significant 0's and the first 1 unchanged and then replacing 1's with 0's and 0's with 1's

“Complement of the complement restores the number”

$$(r^n - 1) - [(r^n - 1) - N] = N$$

$$r^n - [r^n - N] = N$$